

CSULB
CECS174

LabW4

For best result write your answer and then check it using Python IDE

This lab has 4 questions

1- Given the following assignments:

i1 = 2
i2 = 5
i3 = -3
d1 = 2.0
d2 = 5.0
d3 = -0.5

Evaluate each of the following Python expressions.

Expression	Expected value	Evaluation using Python IDE
a. i1 + i2	7	2 + 5 = 7
b. i1 / i2	0.4	2 / 5 = 0.4
c. i1 // i2	0.4	2 / 5 = 0.4
d. i2 / i1	2.5	5 / 2 = 2.5
e. i2 // i1	2.5	5 / 2 = 2.5
f. i1 * i3	-6	2 * (-3) = -6
g. d1 + d2	7	2.0 + 5.0 = 7.0
h. d1 / d2	0.4	2.0 / 5.0 = 0.4
i. d2 / d1	2.5	5.0 / 2.0 = 2.5
j. d3 * d1	-1	-0.5 * 2.0 = -1.0
k. d1 + i2	7	2.0 + 5 = 7.0
l. i1 / d2	0.4	2 / 5.0 = 0.4
m. d2 / i1	2.5	5.0 / 2 = 2.5
n. i2 / d1	2.5	5 / 2.0 = 2.5
o. i1/i2*d1	0.8	2/5 * 2.0 = 0.8
p. d1*i1/i2	0.8	2.0 * 2/5 = 0.8
q. d1/d2*i1	0.8	2.0/5.0 * 2 = 0.8
r. i1*d1/d2	0.8	2 * 2.0/5.0 = 0.8
s. i2/i1*d1	5	5/2 * 2.0 = 5.0
t. d1*i2/i1	5	2.0 * 5/2 = 5.0
u. d2/d1*i1	5	5.0/2.0 * 2 = 5.0
v. i1*d2/d1	5	2 * 5.0/2.0 = 5.0

2- Given the following assignments:

```
i1 = 2
i2 = 5
i3 = -3
d1 = 2.0
d2 = 5.0
d3 = -0.5
```

Evaluate each of the following Python expressions.

Expression	Expected value	Using Python
a. $i1 + (i2 * i3)$	-13	$2 + (5 * -3) = -13$
b. $i1 * (i2 + i3)$	4	$2 * (5 + (-3)) = 4$
c. $i1 / (i2 + i3)$	1	$2 / (5 + (-3)) = 1$
d. $i1 // (i2 + i3)$	1	$2 / (5 + (-3)) = 1$
e. $i1 / i2 + i3$	-2.6	$2 / 5 + (-3) = -2.6$
f. $i1 // i2 + i3$	-2.6	$2 / 5 + (-3) = -2.6$
g. $3 + 4 + 5 / 3$	8.666666666666667	8.666666666666667
h. $3 + 4 + 5 // 3$	8.666666666666667	8.666666666666667
i. $(3 + 4 + 5) / 3$	4	4
j. $(3 + 4 + 5) // 3$	4	4
k. $d1 + (d2 * d3)$	-0.5	$2.0 + (5.0 * -0.5) = -0.5$
l. $d1 + d2 * d3$	-0.5	$2.0 + 5.0 * -0.5 = -0.5$
m. $d1 / d2 - d3$	0.9	$2.0 / 5.0 - (-0.5) = 0.9$
n. $d1 / (d2 - d3)$	0.3636363636	$2.0 / (5.0 - (-0.5)) = 0.36363636$
o. $d1 + d2 + d3 / 3$	6.83333333	$2.0 + 5.0 + (-0.5) / 3 = 6.833333$
p. $(d1 + d2 + d3) / 3$	2.1666666666	$(2.0 + 5.0 + (-0.5)) / 3 = 2.1666$
q. $d1 + d2 + (d3 / 3)$	6.833333333	$2.0 + 5.0 + (-0.5/3) = 6.833333$
r. $3 * (d1 + d2) * (d1 - d3)$	52.5	$3*(2.0+5.0)*(2.0-(-0.5)) = 52.5$

3- Write the shortest way to express each of the following

- a. $x = x + 1$ $x += 1$
- b. $x = x / 2$ $x /= 2$
- c. $x = x - 1$ $x -= 1$
- d. $x = x + y$ $x += y$
- e. $x = x - (y + 7)$ $x -= y+7$
- f. $x = 2 * x$ $x *= 2$
- g. $\text{number_of_seconds_in_my_age} = \text{number_of_seconds_in_my_age} + 2 * \text{someVal}$
 $\text{number_of_seconds_in_my_age} += 2*\text{someVal}$

- 4- Consider the following program which contains some errors. You may assume that the comments within the program accurately describe the program's intended behavior. Please note the ; will allow me to use two statements in one line

Get two numbers from the user

n1 = int (input()) ; n2 = int (input()) #1

Compute sum of the two numbers

print(n1 + n2) #2

Compute average of the two numbers

print(n1+n2/2) #3 There is an logical error because it's calculating, but not the average

Assign some variables

d1=d2=0 #4 There is an error because of double '='

Compute a quotient

print(n1/d1) #5

Compute a product

n1*n2=d1 #6

Print result

print(d1) #7